

Algebra II Trigonometry Cheatsheet

Module 9

Angle Basics

$$180^\circ = \pi \text{ radians}, \quad 360^\circ = 2\pi \text{ radians}$$

$$\theta^\circ = \theta \cdot \frac{\pi}{180}, \quad \theta \text{ radians} = \theta \cdot \frac{180^\circ}{\pi}$$

Special Values

$$\sin\left(\frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3}\right) = \left(\frac{1}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{2}\right)$$

$$\cos\left(\frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3}\right) = \left(\frac{\sqrt{3}}{2}, \frac{\sqrt{2}}{2}, \frac{1}{2}\right)$$

$$\tan\left(\frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3}\right) = \left(\frac{\sqrt{3}}{3}, 1, \sqrt{3}\right)$$

Quadrant signs:

$$\text{QI all,} \quad \text{QII sin,} \quad \text{QIII tan,} \quad \text{QIV cos}$$

Core Identities

$$\sin^2 x + \cos^2 x = 1$$

$$1 + \tan^2 x = \sec^2 x$$

$$1 + \cot^2 x = \csc^2 x$$

$$\tan x = \frac{\sin x}{\cos x}, \quad \cot x = \frac{\cos x}{\sin x}$$

Symmetry and Cofunctions

$$\sin(-x) = -\sin x, \quad \cos(-x) = \cos x, \quad \tan(-x) = -\tan x$$

$$\sin\left(\frac{\pi}{2} - x\right) = \cos x, \quad \cos\left(\frac{\pi}{2} - x\right) = \sin x$$

$$\tan\left(\frac{\pi}{2} - x\right) = \cot x$$

Angle Formulas

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$$

Double and Half Angles

$$\sin(2x) = 2 \sin x \cos x$$

$$\cos(2x) = \cos^2 x - \sin^2 x = 2 \cos^2 x - 1 = 1 - 2 \sin^2 x$$

$$\tan(2x) = \frac{2 \tan x}{1 - \tan^2 x}$$

$$\sin\left(\frac{x}{2}\right) = \pm \sqrt{\frac{1 - \cos x}{2}}, \quad \cos\left(\frac{x}{2}\right) = \pm \sqrt{\frac{1 + \cos x}{2}}$$

The sign depends on the quadrant of $\frac{x}{2}$.

Graphing

$$y = a \sin(b(x - h)) + k$$

$$y = a \cos(b(x - h)) + k$$

$$\text{amplitude} = |a|, \quad \text{period} = \frac{2\pi}{|b|}$$

$$\text{midline } y = k, \quad \text{phase shift } h$$

For tangent:

$$y = a \tan(b(x - h)) + k, \quad \text{period} = \frac{\pi}{|b|}$$

Inverse Trig

$$\sin^{-1} x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

$$\cos^{-1} x \in [0, \pi]$$

$$\tan^{-1} x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

Solving Trig Equations

- Isolate the trig expression.
- Find reference angle.
- Use quadrant signs.
- List all solutions in the requested interval.